

Qualification Pack



Solar Water Pumping Junior Engineer

QP Code: SGJ/Q0112

Version: 4.0

NSQF Level: 5

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi - 110021 || email:prem@sscgj.in

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SGJ/Q0112: Solar Water Pumping Junior Engineer

Brief Job Description

A Solar Water Pumping Junior Engineer specializes in the design, installation and commissioning of the solar PV power plant, its quality assurance and Health & Safety issues. He/she also specializes in designing, installation and commissioning of solar water pumping systems.

Personal Attributes

This job requires the individual to oversee the complete installation, testing and commissioning of solar PV power plants and ensure no accidents occur at site, so diligence and hardworking are desired attributes for individuals performing this role. The job holder must also demonstrate strong work ethics, an ability to communicate courteously with co-workers and sub-ordinates

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N0109: Prepare a site feasibility study report](#)
2. [SGJ/N0146: Design of solar PV power plant](#)
3. [SGJ/N0132: Installation and Commissioning of solar PV power plant](#)
4. [SGJ/N0133: Quality Assurance of Solar PV power plant & components](#)
5. [SGJ/N0106: Maintain Personal Health & Safety at project site](#)
6. [SGJ/N0134: Design, Installation and Commissioning of Solar Water Pumping System](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	PV System Installation Engineer
Country	India
NSQF Level	5

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Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7421.1403
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) ((Electrical/Electronics) or equivalent) with NA of experience OR 12th grade pass with 1 year NTC/ NAC (12th Grade pass with 1 year of NTC/NAC in relevant trade) with 1-2 Years of experience 1.5 years of relevant experience in renewable energy/ power sector OR Completed 3 year diploma after 10th (Any stream of Engineering) with 1-2 Years of experience 1.5 years of relevant experience in renewable energy/ power sector OR Previous relevant Qualification of NSQF Level (Previous relevant qualification of level 4.5) with 1-2 Years of experience 1.5 years of relevant experience in renewable energy/ power sector OR OR Previous relevant Qualification of NSQF Level (Previous relevant qualification of level 4 (Solar PV Installer (Suryamitra)) with 3 Years of experience 3 years of relevant experience in renewable energy/ power sector
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	29/05/2027
NSQC Approval Date	30/05/2024
Version	4.0
Reference code on NQR	QG-05-ES-02631-2024-V2-SCGJ
NQR Version	2

Remarks:

Total 540 hours i.e. 18 Credits (Theory : 225 hours + Practical : 225 hours + OJT : 90 hours)

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SGJ/N0109: Prepare a site feasibility study report

Description

This unit is about the key steps involved in developing a site feasibility report for rooftop solar PV system along with assessing client requirement

Scope

The scope covers the following :

- Introduction to Solar PV Sector in India
- Prepare a site Feasibility Study Report

Elements and Performance Criteria

Introduction to Solar PV Sector in India

To be competent, the user/individual on the job must be able to:

- PC1.** discuss and provide an overview of solar PV technology
- PC2.** discuss and provide an overview of ground mounted and rooftop solar sector
- PC3.** discuss types of ground mounted PV Power Plants and their working principles
- PC4.** discuss various types of Rooftop Solar PV Power Plant and working principles
- PC5.** discuss solar PV water pumping system technology and market
- PC6.** discuss and provide an overview of National and State specific Solar PV water pumping related policies (e.g. PM KUSUM) and regulations
- PC7.** discuss and provide an overview of available subsidies and financing options of solar water pumping system
- PC8.** show the function of any solar PV system including solar water pumping system
- PC9.** demonstrate know how of installation of solar water pump.
- PC10.** show how financing for solar water pumping system can be arranged and how subsidies can be availed

Prepare a site Feasibility Study Report

To be competent, the user/individual on the job must be able to:

- PC11.** discuss how to identify optimum location for installations
- PC12.** discuss different types of Roofs and their structural load profile
- PC13.** discuss how to assess the site level pre-requisites for solar panel installation on both ground and roof
- PC14.** discuss how to analyse type of mounting to be constructed and place of mounting as per client requirement
- PC15.** discuss how to check for any shading obstacles
- PC16.** discuss how to prepare a site map of the location where installation has to be carried out
- PC17.** discuss how to assess the load to be run on solar PV power plant and prepare a load profile
- PC18.** discuss how to estimate the capacity of solar PV power plant

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- PC19.** discuss how to decide on battery backup as per grid availability, loads and client expectation
- PC20.** discuss how to assess or obtain the site specific major parameters of solar resource data like Global Horizontal Irradiance (GHI), Diffused Normal Irradiance (DNI), Temperature and Wind
- PC21.** discuss how to perform shading analysis using simulation tools
- PC22.** discuss how to identify the risks associated with the specific solar project
- PC23.** discuss how to prepare a site Feasibility Study Report using specialized software like PV*SOL®, PVsyst, SketchUp etc
- PC24.** show how to perform simple calculations to illustrate the fundamental concepts of power and energy
- PC25.** show how the movement of sun affects the performance of the solar power plant
- PC26.** show how to obtain site specific resource data like GHI, DNI etc
- PC27.** show how to estimate the energy generated from the solar PV power plant using solar design software like PV*SOL®, PVsyst, etc..
- PC28.** show how to prepare a site feasibility report

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys policies on: incentives, personnel management
- KU2.** companys code of conduct
- KU3.** importance of individuals role in the work flow
- KU4.** companys documentation policy
- KU5.** companys installation policy
- KU6.** companys customer support policy
- KU7.** simple calculations to derive the power and energy received from solar radiation in a given area
- KU8.** solar resource assessment including direct normal irradiation, diffuse horizontal irradiation, global horizontal irradiation and albedo
- KU9.** ground based measurement and satellite derived data
- KU10.** building orientation
- KU11.** types of roofs and suggestive mounting structure for that specific roof
- KU12.** basic concepts of Trigonometry and coordinate geometry
- KU13.** effect on array output of current and voltage based on series / parallel connections of modules, tilt angle, orientation and shading
- KU14.** effect of shading on the system using tools like Solar path finder and softwares like PV*SOL, PVsyst, etc.
- KU15.** cabling route and length of cable required
- KU16.** different types of tracking systems
- KU17.** simulation software
- KU18.** risks associated with the solar project
- KU19.** synchronization with diesel generators

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read warnings, instructions and other text material on product labels, components etc.
- GS2.** express statements for information clearly so that customer can hear and understand
- GS3.** follow code of conduct
- GS4.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS5.** interact with client to analyse client exact requirement
- GS6.** interact with workers in order to coordinate work processes
- GS7.** participate in and understand main points of simple discussions with customer
- GS8.** respond properly to any query of the customer

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Solar PV Sector in India</i>	21	15	-	-
PC1. discuss and provide an overview of solar PV technology	3	-	-	-
PC2. discuss and provide an overview of ground mounted and rooftop solar sector	3	-	-	-
PC3. discuss types of ground mounted PV Power Plants and their working principles	3	-	-	-
PC4. discuss various types of Rooftop Solar PV Power Plant and working principles	3	-	-	-
PC5. discuss solar PV water pumping system technology and market	3	-	-	-
PC6. discuss and provide an overview of National and State specific Solar PV water pumping related policies (e.g. PM KUSUM) and regulations	3	-	-	-
PC7. discuss and provide an overview of available subsidies and financing options of solar water pumping system	3	-	-	-
PC8. show the function of any solar PV system including solar water pumping system	-	5	-	-
PC9. demonstrate know how of installation of solar water pump.	-	5	-	-
PC10. show how financing for solar water pumping system can be arranged and how subsidies can be availed	-	5	-	-
<i>Prepare a site Feasibility Study Report</i>	39	25	-	-
PC11. discuss how to identify optimum location for installations	3	-	-	-
PC12. discuss different types of Roofs and their structural load profile	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss how to assess the site level pre-requisites for solar panel installation on both ground and roof	3	-	-	-
PC14. discuss how to analyse type of mounting to be constructed and place of mounting as per client requirement	3	-	-	-
PC15. discuss how to check for any shading obstacles	3	-	-	-
PC16. discuss how to prepare a site map of the location where installation has to be carried out	3	-	-	-
PC17. discuss how to assess the load to be run on solar PV power plant and prepare a load profile	3	-	-	-
PC18. discuss how to estimate the capacity of solar PV power plant	3	-	-	-
PC19. discuss how to decide on battery backup as per grid availability, loads and client expectation	3	-	-	-
PC20. discuss how to assess or obtain the site specific major parameters of solar resource data like Global Horizontal Irradiance (GHI), Diffused Normal Irradiance (DNI), Temperature and Wind	3	-	-	-
PC21. discuss how to perform shading analysis using simulation tools	3	-	-	-
PC22. discuss how to identify the risks associated with the specific solar project	3	-	-	-
PC23. discuss how to prepare a site Feasibility Study Report using specialized software like PV*SOL®, PVsyst, SketchUp etc	3	-	-	-
PC24. show how to perform simple calculations to illustrate the fundamental concepts of power and energy	-	5	-	-
PC25. show how the movement of sun affects the performance of the solar power plant	-	5	-	-
PC26. show how to obtain site specific resource data like GHI, DNI etc	-	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. show how to estimate the energy generated from the solar PV power plant using solar design software like PV*SOL®, PVsyst, etc..	-	5	-	-
PC28. show how to prepare a site feasibility report	-	5	-	-
NOS Total	60	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0109
NOS Name	Prepare a site feasibility study report
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Solar Site Survey
NSQF Level	5
Credits	2
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

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SGJ/N0146: Design of solar PV power plant

Description

This unit is about reviewing of structural design, preparing the electrical design of solar PV power plant

Scope

The scope covers the following :

- Design of solar PV power plant

Elements and Performance Criteria

Design of solar PV power plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss and show how to review and correctly interpret the mounting structure and foundation design & drawings
- PC2.** discuss and show how to review the overall structural layout of the solar PV power plant
- PC3.** discuss and show how to select solar module technology and size, based on analysis of cost, power output, quality, climatic conditions of the site, global and diffused irradiance ratio at the site, etc
- PC4.** discuss and show how to calculate the total numbers of modules based on the total capacity of the plant and the capacity of selected modules
- PC5.** discuss to prepare the earthing design of solar module arrays
- PC6.** discuss how to select inverter, based on compatibility with module technology, compliance with grid code and other applicable regulations, reliability, system availability, serviceability, quality, cost
- PC7.** discuss how to decide on specifications of the inverter to power the AC loads in the building, in rooftop system
- PC8.** discuss and show how to decide on number of inverters to be used based on the capacity and specifications of the inverter selected
- PC9.** discuss to finalize the inverter layout and inverter locations on the basis of total capacity
- PC10.** discuss to prepare the earthing design of inverters.
- PC11.** discuss and show how to calculate number of modules in a string based on the input voltage and MPPT voltage range of the inverter.
- PC12.** discuss and show how to estimate number of strings connected to a combiner box based on minimum run of DC connecting cables to minimized DC losses.
- PC13.** discuss and show how to calculate the number of combiner boxes connected to one panel of the inverter based on the input current rating of the inverter.
- PC14.** discuss how to finalize the inter row distance between the solar modules on the basis of minimum inter row shading, adequate space for cleaning and maintenance of solar modules and the tilted to south at an angle that optimizes the annual energy yield.
- PC15.** discuss and show how to decide the battery storage capacity (AH) based on the number of days autonomy required (KWH/WH) and the depth of discharge of the battery bank.

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- PC16.** discuss and show how to decide on specification of charge controller/ inverter to the control the overcharging/discharging of batteries.
- PC17.** discuss and show how to prepare energy generation report using simulation software.
- PC18.** discuss how to specify DC cabling material, size, type of PVC for cables connecting modules, junction boxes to the combiner boxes and combiner boxes to the inverter panels etc.
- PC19.** discuss to prepare the specification of DC connectors (plugs and sockets) to be used.
- PC20.** discuss how to prepare the design specifications for junction boxes/combiner including IP number.
- PC21.** discuss to prepare the specifications for disconnects/switches
- PC22.** discuss to prepare islanding facility for grid connected power plant, in case of non-availability of grid.
- PC23.** discuss to protect incorrect polarity, over-voltage and overload for the DC cables.
- PC24.** discuss to decide on the specifications for the charge controller/ inverter to control the overcharging/discharging of the batteries.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** government/corporate policies and guidelines on: workplace safety, identification and mitigation of safety hazards, work procedures and guidelines for working at height
- KU2.** document information using appropriate corporate forms
- KU3.** obtain authorization from specified field safety officer and supervisor
- KU4.** legislative, organization, site requirements and procedures
- KU5.** efficiency, cost and typical specifications, functioning and operating principle of different types of solar PV plants, commercially available PV cells and modules, inverters, transformers, charge controllers, battery, mounting structures, cables, junction boxes and other components
- KU6.** site survey reports , availability of shadow free space for installation of solar power plant
- KU7.** the survey equipment and the methodology of survey
- KU8.** electrical designs for the module/ inverters and balance of system
- KU9.** solar irradiation including GHI, DHI and DNI
- KU10.** mechanical and electrical features necessary for the long life of the PV Power Plant under a wide range of operating conditions
- KU11.** solar PV power plant design software such as PVSYST and PV*SOL etc.
- KU12.** energy simulation report and its parameters and effect on solar PV plants

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare documentation as per relevant industry standards
- GS2.** read vernacular/English language

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- GS3.** how to read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** how to read from different sources- books, screens in machines and signage
- GS5.** the various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with peers, superiors and sub-ordinates
- GS10.** follow organization rule-based decision making process
- GS11.** analyze critical points in day to day tasks and identify control measures to solve the issue
- GS12.** handle issues in case the superior is not available with clear choice of procedures in familiar contexts (as per the authority matrix defined by the organisation)
- GS13.** plan and organize work to meet deadlines
- GS14.** work constructively and collaboratively with others
- GS15.** support the superiors in scheduling tasks
- GS16.** follow organisation code of conduct
- GS17.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS18.** recognize problems and provide solutions using a range of cognitive and practical skills
- GS19.** approach relevant authority when required
- GS20.** apply knowledge of facts, principles and processes to select the right course of action to perform tasks
- GS21.** use reasoning skills to identify and resolve basic problems
- GS22.** use intuition to detect any potential problems which could arise during operations
- GS23.** use acquired knowledge of the process for identifying and handling issues

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design of solar PV power plant</i>	67	33	-	-
PC1. discuss and show how to review and correctly interpret the mounting structure and foundation design & drawings	2	3	-	-
PC2. discuss and show how to review the overall structural layout of the solar PV power plant	2	3	-	-
PC3. discuss and show how to select solar module technology and size, based on analysis of cost, power output, quality, climatic conditions of the site, global and diffused irradiance ratio at the site, etc	3	3	-	-
PC4. discuss and show how to calculate the total numbers of modules based on the total capacity of the plant and the capacity of selected modules	3	3	-	-
PC5. discuss to prepare the earthing design of solar module arrays	3	-	-	-
PC6. discuss how to select inverter, based on compatibility with module technology, compliance with grid code and other applicable regulations, reliability, system availability, serviceability, quality, cost	3	-	-	-
PC7. discuss how to decide on specifications of the inverter to power the AC loads in the building, in rooftop system	3	-	-	-
PC8. discuss and show how to decide on number of inverters to be used based on the capacity and specifications of the inverter selected	3	3	-	-
PC9. discuss to finalize the inverter layout and inverter locations on the basis of total capacity	3	-	-	-
PC10. discuss to prepare the earthing design of inverters.	3	-	-	-
PC11. discuss and show how to calculate number of modules in a string based on the input voltage and MPPT voltage range of the inverter.	3	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. discuss and show how to estimate number of strings connected to a combiner box based on minimum run of DC connecting cables to minimized DC losses.	3	3	-	-
PC13. discuss and show how to calculate the number of combiner boxes connected to one panel of the inverter based on the input current rating of the inverter.	3	3	-	-
PC14. discuss how to finalize the inter row distance between the solar modules on the basis of minimum inter row shading, adequate space for cleaning and maintenance of solar modules and the tilted to south at an angle that optimizes the annual energy yield.	3	-	-	-
PC15. discuss and show how to decide the battery storage capacity (AH) based on the number of days autonomy required (KWH/WH) and the depth of discharge of the battery bank.	3	3	-	-
PC16. discuss and show how to decide on specification of charge controller/ inverter to the control the overcharging/discharging of batteries.	3	3	-	-
PC17. discuss and show how to prepare energy generation report using simulation software.	3	3	-	-
PC18. discuss how to specify DC cabling material, size, type of PVC for cables connecting modules, junction boxes to the combiner boxes and combiner boxes to the inverter panels etc.	3	-	-	-
PC19. discuss to prepare the specification of DC connectors (plugs and sockets) to be used.	3	-	-	-
PC20. discuss how to prepare the design specifications for junction boxes/combiner including IP number.	2	-	-	-
PC21. discuss to prepare the specifications for disconnects/switches	3	-	-	-
PC22. discuss to prepare islanding facility for grid connected power plant, in case of non- availability of grid.	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. discuss to protect incorrect polarity, over-voltage and overload for the DC cables.	2	-	-	-
PC24. discuss to decide on the specifications for the charge controller/ inverter to control the overcharging/discharging of the batteries.	3	-	-	-
NOS Total	67	33	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0146
NOS Name	Design of solar PV power plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	PV System Installation Engineer
NSQF Level	5
Credits	4
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

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SGJ/N0132: Installation and Commissioning of solar PV power plant

Description

This unit is about understanding the process involved during installation and commissioning of solar PV power plant

Scope

The scope covers the following :

- Installation and commissioning of solar PV power plant

Elements and Performance Criteria

Installation and commissioning of solar PV power plant

To be competent, the user/individual on the job must be able to:

- PC1.** discuss and show how to interpret the design and detailed drawings of the civil, mechanical and electrical works to be carried out at site
- PC2.** discuss how to arrange for tools and consumable required for installation.
- PC3.** discuss how to follow the schedule for each of the civil and mechanical construction activity
- PC4.** discuss and show how to manage the schedule for installation of modules, inverters, transformers, power protection devices, lightning arresters, earthing systems, etc. and ensure installation as per the design documents
- PC5.** discuss how to check cables for continuity and manage the installation of communication and storage system with SCADA facility/ any monitoring system
- PC6.** discuss and show how to ensure installation of battery banks, if required
- PC7.** discuss and show how to prepare, review and report progress on daily basis to the site in-charge for further action
- PC8.** discuss and show how to visually inspect the plant and get pre connection connectivity and conductivity test done
- PC9.** discuss and show how to verify system grounding and get the insulation resistance measured
- PC10.** discuss as per design Confirm that electrical protections, disconnection and other provisions are fulfilled as per design documents.
- PC11.** discuss and show how to get the DC voltage and current test done for each of the module strings
- PC12.** discuss and show how to measure and record all relevant parameters of energy storage system if present
- PC13.** discuss how to ensure calibration of SCADA/any monitoring system
- PC14.** discuss and show how to prepare inspection report and forward to site-in charge for further action
- PC15.** discuss how to the getting the clearance from electricity inspector, initiate startup procedures as per manufacturer's instructions

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- PC16.** discuss and show how to monitor the energy readings and voltages at regular intervals on start up
- PC17.** discuss on record and report any anomalous condition to the site in-charge for further action
- PC18.** discuss and show how to prepare as-built drawings and document design changes, if any
- PC19.** explain how to perform trouble shooting of a Solar PV system, during commissioning and during maintenance
- PC20.** explain Field parameter settings of Inverter
- PC21.** discuss common challenges in installation and troubleshooting those.
- PC22.** show how to ensure the installation of cables between different components as per design documents
- PC23.** show how to prepare inspection report.
- PC24.** show how to identify materials which can be replaced by environment friendly substitutes.
- PC25.** show how to identify processes where material and resource utilization can be optimized.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** government/corporate policies and guidelines on: workplace safety, identification and mitigation of safety hazards, work procedures and guidelines for working at height
- KU2.** document information using appropriate corporate forms
- KU3.** obtain authorization from specified field safety officer and supervisor
- KU4.** legislative, organization, site requirements and procedures
- KU5.** definition of the terms: energy and power, cell, module, string, array, mono-crystalline, poly-crystalline, amorphous silicon
- KU6.** efficiency, cost and typical specifications, functioning and operating principle of different types of solar photovoltaic plants, commercially available PV modules, inverters, charge controllers, battery, mounting structures, cables, junction boxes and other components
- KU7.** fundamentals of solar resource like GHI, DNI etc. and measurement of solar irradiance with a pyranometer
- KU8.** effect on array output of current and voltage based on series / parallel connections of modules, tilt angle, orientation and shading
- KU9.** simple calculations to derive the power and energy received from solar radiation in a given area
- KU10.** mechanical and electrical features necessary for the long life of the PV power plant under a wide range of operating conditions
- KU11.** site survey reports, availability of shadow free space for installation of solar power plant
- KU12.** structural designs and foundation plans of rooftop and ground mount systems
- KU13.** dos and don'ts of DC wiring and installation of other electrical equipment
- KU14.** dos and don'ts of material handling and storage
- KU15.** basic functioning and operation of different types of inverters and other electrical equipment
- KU16.** connection of solar power plant to LT panel and switchover along with precautions

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- KU17.** testing and commissioning activities and their interpretation- continuity of wiring, earthing, polarity check, insulation, voltage drop
- KU18.** measurement of losses in a PV systems at different points and interpretation of results
- KU19.** typical faults, their causes and resolution for all system components
- KU20.** grid codes and other regulatory provisions
- KU21.** methods of optimum utilization of material and resources

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare documentation as per relevant industry standards
- GS2.** read vernacular/English language
- GS3.** read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** read and interpret data from various sources
- GS5.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with peers, superiors and sub-ordinates
- GS10.** follow organization rule-based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize service work to meet deadlines
- GS13.** organize raw materials and packaging materials required for site survey
- GS14.** plan to utilize time and equipment's effectively
- GS15.** work constructively and collaboratively with others
- GS16.** follow code of conduct
- GS17.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS18.** recognize problems and provide solutions using a range of cognitive and practical skills
- GS19.** approach relevant authority when required
- GS20.** apply knowledge of facts, principles and processes to select the right course of action to perform tasks
- GS21.** use reasoning skills to identify and resolve basic problems
- GS22.** use intuition to detect any potential problems which could arise during operations
- GS23.** use acquired knowledge of the process for identifying and handling issues
- GS24.** record data relevant to material and resource usage at workplace

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation and commissioning of solar PV power plant</i>	67	33	-	-
PC1. discuss and show how to interpret the design and detailed drawings of the civil, mechanical and electrical works to be carried out at site	2	3	-	-
PC2. discuss how to arrange for tools and consumable required for installation.	3	-	-	-
PC3. discuss how to follow the schedule for each of the civil and mechanical construction activity	3	-	-	-
PC4. discuss and show how to manage the schedule for installation of modules, inverters, transformers, power protection devices, lightning arresters, earthing systems, etc. and ensure installation as per the design documents	3	3	-	-
PC5. discuss how to check cables for continuity and manage the installation of communication and storage system with SCADA facility/ any monitoring system	3	-	-	-
PC6. discuss and show how to ensure installation of battery banks, if required	3	3	-	-
PC7. discuss and show how to prepare, review and report progress on daily basis to the site in-charge for further action	3	3	-	-
PC8. discuss and show how to visually inspect the plant and get pre connection connectivity and conductivity test done	3	3	-	-
PC9. discuss and show how to verify system grounding and get the insulation resistance measured	2	3	-	-
PC10. discuss as per design Confirm that electrical protections, disconnection and other provisions are fulfilled as per design documents.	3	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. discuss and show how to get the DC voltage and current test done for each of the module strings	3	3	-	-
PC12. discuss and show how to measure and record all relevant parameters of energy storage system if present	2	3	-	-
PC13. discuss how to ensure calibration of SCADA/any monitoring system	2	-	-	-
PC14. discuss and show how to prepare inspection report and forward to site-in charge for further action	3	3	-	-
PC15. discuss how to the getting the clearance from electricity inspector, initiate startup procedures as per manufacturer's instructions	3	-	-	-
PC16. discuss and show how to monitor the energy readings and voltages at regular intervals on start up	2	3	-	-
PC17. discuss on record and report any anomalous condition to the site in-charge for further action	3	-	-	-
PC18. discuss and show how to prepare as-built drawings and document design changes, if any	3	3	-	-
PC19. explain how to perform trouble shooting of a Solar PV system, during commissioning and during maintenance	2	-	-	-
PC20. explain Field parameter settings of Inverter	2	-	-	-
PC21. discuss common challenges in installation and troubleshooting those.	2	-	-	-
PC22. show how to ensure the installation of cables between different components as per design documents	3	-	-	-
PC23. show how to prepare inspection report.	3	-	-	-
PC24. show how to identify materials which can be replaced by environment friendly substitutes.	3	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. show how to identify processes where material and resource utilization can be optimized.	3	-	-	-
NOS Total	67	33	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0132
NOS Name	Installation and Commissioning of solar PV power plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	PV System Installation Engineer
NSQF Level	5
Credits	3
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

SGJ/N0133: Quality Assurance of Solar PV power plant & components

Description

This unit provides an overview about how to check modules and inverters as per relevant standards, inspect key components including installation of modules, inverters, protection devices and entire PV system while ensuring that documentations of all components are in order.

Scope

The scope covers the following :

- Quality Assurance of Solar PV power plant & components

Elements and Performance Criteria

Quality Assurance of Solar PV power plant & components

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to check modules earmarked for power plant using a random selection as per relevant IS/IEC standards
- PC2.** discuss how to collect documentation related to each and every equipment and submit to site in-charge
- PC3.** discuss how to ensure proper delivery/off-load of solar equipment
- PC4.** discuss how to ensure to check all the material and equipment received at site for any physical damage
- PC5.** discuss how to ensure specifications of the equipment and components match with what has been ordered
- PC6.** discuss to ensure all warranty documents by manufacturers are properly signed and are in order
- PC7.** discuss how to inspect the foundations of structures and verify cable routes and specifications as per design documents
- PC8.** discuss how to inspect module installation, cable terminations while ensuring tightness
- PC9.** discuss how to inspect the installation of inverters, protection devices and systems
- PC10.** discuss how to carry out thermography of doubtful strings and modules to know the defects
- PC11.** discuss how to collect and compile conformity, warranty documentation, performance guarantees, calibration certificates and any other relevant documentation and handover to site incharge, certificates
- PC12.** show how to visit of the manufacturing facility of module manufacturer and inverter supplier and witness testing of a few inverters
- PC13.** show how to inspect the foundations of structures and verify cable routes and specifications as per design documents
- PC14.** show how to inspect module installation, cable terminations, installed inverters and protection devices and systems
- PC15.** show how to perform visual inspection of the plant to find out defects and deficiencies

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- PC16.** show how to measure and record the circuit voltage and short circuit current of all the module strings and compare that with design values
- PC17.** show how to carry out performance ratio test by continuous operation of the plant as per the industry norms and compare with designed values
- PC18.** show how to implement cost-effective approaches to manage and maintain solar-powered systems

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** government/corporate policies and guidelines on: workplace safety, identification and mitigation of safety hazards, work procedures and guidelines for working at height
- KU2.** document information using appropriate corporate forms
- KU3.** obtain authorization from specified field safety officer and supervisor
- KU4.** legislative, organization, site requirements and procedures
- KU5.** relevant IEC/IS standards and MNRE guidelines
- KU6.** all test and inspection documentation of various components
- KU7.** testing and commissioning activities and their interpretation- continuity of wiring, earthing, polarity check, insulation, voltage drop
- KU8.** measurement of losses in a PV systems at different interconnections and interpretation of the results
- KU9.** typical faults, their causes and resolution for all system components
- KU10.** IV curve and performance ratio tests

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare documentation as per relevant industry standards
- GS2.** read vernacular/English language
- GS3.** read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** ability to read from different sources- books, screens in machines and signage
- GS5.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with peers, superiors and sub-ordinates
- GS10.** follow organization rule-based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize service work to meet deadlines

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- GS13.** organize raw materials and packaging materials required for site survey
- GS14.** plan to utilize time and equipment's effectively
- GS15.** work constructively and collaboratively with others
- GS16.** follow code of conduct
- GS17.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS18.** recognize problems and provide solutions using a range of cognitive and practical skills
- GS19.** approach relevant authority when required
- GS20.** apply knowledge of facts, principles and processes to select the right course of action to perform tasks
- GS21.** use reasoning skills to identify and resolve basic problems
- GS22.** use intuition to detect any potential problems which could arise during operations
- GS23.** use acquired knowledge of the process for identifying and handling issues

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Quality Assurance of Solar PV power plant & components</i>	25	25	-	-
PC1. discuss how to check modules earmarked for power plant using a random selection as per relevant IS/IEC standards	2	-	-	-
PC2. discuss how to collect documentation related to each and every equipment and submit to site in-charge	2	-	-	-
PC3. discuss how to ensure proper delivery/off-load of solar equipment	2	-	-	-
PC4. discuss how to ensure to check all the material and equipment received at site for any physical damage	2	-	-	-
PC5. discuss how to ensure specifications of the equipment and components match with what has been ordered	2	-	-	-
PC6. discuss to ensure all warranty documents by manufacturers are properly signed and are in order	2	-	-	-
PC7. discuss how to inspect the foundations of structures and verify cable routes and specifications as per design documents	2	-	-	-
PC8. discuss how to inspect module installation, cable terminations while ensuring tightness	2	-	-	-
PC9. discuss how to inspect the installation of inverters, protection devices and systems	3	-	-	-
PC10. discuss how to carry out thermography of doubtful strings and modules to know the defects	3	-	-	-
PC11. discuss how to collect and compile conformity, warranty documentation, performance guarantees, calibration certificates and any other relevant documentation and handover to site incharge, certificates	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. show how to visit of the manufacturing facility of module manufacturer and inverter supplier and witness testing of a few inverters	-	4	-	-
PC13. show how to inspect the foundations of structures and verify cable routes and specifications as per design documents	-	4	-	-
PC14. show how to inspect module installation, cable terminations, installed inverters and protection devices and systems	-	4	-	-
PC15. show how to perform visual inspection of the plant to find out defects and deficiencies	-	4	-	-
PC16. show how to measure and record the circuit voltage and short circuit current of all the module strings and compare that with design values	-	4	-	-
PC17. show how to carry out performance ratio test by continuous operation of the plant as per the industry norms and compare with designed values	-	5	-	-
PC18. show how to implement cost-effective approaches to manage and maintain solar-powered systems	-	-	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0133
NOS Name	Quality Assurance of Solar PV power plant & components
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	PV System Installation Engineer
NSQF Level	5
Credits	2
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

SGJ/N0106: Maintain Personal Health & Safety at project site

Description

This unit is about maintaining health, safety and hygiene at workplace.

Scope

The scope covers the following :

- Adopt safe practices at workplace
- Follow emergencies, rescue and first aid procedures
- Follow good housekeeping practices and infection control guidelines

Elements and Performance Criteria

Adopt safe practices at workplace

To be competent, the user/individual on the job must be able to:

- PC1.** explain the requirements for safe work area
- PC2.** identify and report any hazards, risks or breaches in site safety to the appropriate authority
- PC3.** follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk
- PC4.** use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition
- PC5.** follow safe practices when working at height and in confined space
- PC6.** handle all required tools, tackles, materials and equipment safely
- PC7.** identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures
- PC8.** apply ergonomic principles wherever required
- PC9.** use safety signs, labels, charts and notices at workplace
- PC10.** identify work safety procedures and instructions for handling heavy components

Follow emergencies, rescue and first aid procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow emergency and evacuation procedures in case of accidents, fires and natural calamities
- PC12.** use appropriate fire extinguishers for different types of fire
- PC13.** administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.
- PC14.** use correct method to move injured person during an emergency

Follow good housekeeping practices and infection control guidelines

To be competent, the user/individual on the job must be able to:

- PC15.** follow recommended personal hygiene, workplace hygiene and sanitation practices
- PC16.** clean and disinfect all material, tools and supplies before and after use

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- PC17.** report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues
- PC18.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of safety drills
- KU2.** importance of working in clean and safe environment
- KU3.** health and safety roles and responsibilities of relevant personnel within and outside the organization
- KU4.** reporting procedures in case of breaches or hazards for site safety, accidents and emergency situations
- KU5.** basic ergonomic principle
- KU6.** key internal and external source of health and safety information
- KU7.** meaning of hazards, risk and near miss
- KU8.** importance of Personal Protective Equipment required for specific job
- KU9.** forms and classification of hazardous substances
- KU10.** health effect associated with exposure to environmental pollution
- KU11.** housekeeping activities relevant to task
- KU12.** symptoms of infection like fever, cough, swelling and inflammation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data on waste disposal at workplace
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** fill safety formats for near miss, unsafe condition
- GS4.** identify potential safety risk and report to appropriate authority
- GS5.** communicate and collaborate with others to incorporate sustainable practices
- GS6.** interpret general safety guidelines, labels, charts and signage

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adopt safe practices at workplace</i>	13	19	-	-
PC1. explain the requirements for safe work area	2	-	-	-
PC2. identify and report any hazards, risks or breaches in site safety to the appropriate authority	2	3	-	-
PC3. follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk	1	2	-	-
PC4. use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition	2	4	-	-
PC5. follow safe practices when working at height and in confined space	1	1	-	-
PC6. handle all required tools, tackles, materials and equipment safely	1	2	-	-
PC7. identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures	1	2	-	-
PC8. apply ergonomic principles wherever required	1	2	-	-
PC9. use safety signs, labels, charts and notices at workplace	1	1	-	-
PC10. identify work safety procedures and instructions for handling heavy components	1	2	-	-
<i>Follow emergencies, rescue and first aid procedures</i>	4	4	-	-
PC11. follow emergency and evacuation procedures in case of accidents, fires and natural calamities	1	1	-	-
PC12. use appropriate fire extinguishers for different types of fire	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.	1	1	-	-
PC14. use correct method to move injured person during an emergency	1	1	-	-
<i>Follow good housekeeping practices and infection control guidelines</i>	4	6	-	-
PC15. follow recommended personal hygiene, workplace hygiene and sanitation practices	1	1	-	-
PC16. clean and disinfect all material, tools and supplies before and after use	1	1	-	-
PC17. report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues	1	2	-	-
PC18. follow processes specified for disposal of hazardous waste	1	2	-	-
NOS Total	21	29	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0106
NOS Name	Maintain Personal Health & Safety at project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health & Safety , Installation, Solar Panel Installation Technician
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SGJ/N0134: Design, Installation and Commissioning of Solar Water Pumping System

Description

This unit explains about how to perform solar resource assessment, analyse ground water table and decide on the specifications of pumping set and motor, plan the capacity of module, design mounting structure and perform installation and commission of solar pump, various technical aspects of a Pump controller and other components to perform trouble shooting of solar water pumping system. This unit also explain about how to perform periodic maintenance of the water pumping system.

Scope

The scope covers the following :

- Design, installation and commissioning of solar water pumping system

Elements and Performance Criteria

Design, installation and commissioning of solar water pumping system

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to carry out Solar resource assessment
- PC2.** discuss how to decide on the specifications of the pumping set and motor
- PC3.** discuss how to decide on the capacity of PV modules
- PC4.** discuss how to design the plan of mounting structures and foundation
- PC5.** discuss how to oversee the preparation of the foundation for solar module mounting structure and motor pump set.
- PC6.** discuss how to ensure structure is fixed on the foundations
- PC7.** discuss how to oversee the mounting of solar modules; connection of solar module array to pump set in case of DC pumps and installation of inverter (in case of AC pumps).
- PC8.** discuss how to ensure protection system are in place
- PC9.** discuss how to compare the output with design output and take corrective actions, if required
- PC10.** discuss how to ensure connection of the solar module array to motor pump set through a Maximum Power Point Tracker (MPPT) to get maximum power from the array
- PC11.** discuss how to install an inverter after MPPT to convert DC power to AC power in case an AC submersible motor pump set is used
- PC12.** discuss how to operate solar-powered water pump systems effectively and Implement routine maintenance practices for system performance.
- PC13.** discuss how to ensure periodical cleaning of solar module array; periodically ensure tightness of cable connections and periodic maintenance of motor pump set
- PC14.** show how to perform Solar resource assessment
- PC15.** show how to analyze the water usage and level of water table at site
- PC16.** show how to decide on the specifications of the pumping set and motor

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- PC17.** show how to decide on the capacity of PV modules and design the plan of mounting structures and foundation
- PC18.** show how to oversee the preparation of the foundation for solar module mounting structure and motor pump set
- PC19.** show how to ensure that structure is fixed on the foundations
- PC20.** show how to perform inspection and testing of equipment
- PC21.** show how to perform start-up procedures and measure output
- PC22.** show how to compare the output with design output and take corrective actions, if required
- PC23.** show how to install an inverter after MPPT to convert DC power to AC power in case an AC submersible motor pump set is used
- PC24.** show how to assess various technical aspects of a Pump controller to perform trouble shooting of a Solar PV Pumping system
- PC25.** show how to operate solar-powered water pump systems effectively and implement various routine maintenance practices for improved system performance.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's installation policy
- KU2.** company's customer support policy
- KU3.** company's customer support policy
- KU4.** document information using appropriate corporate forms
- KU5.** obtain authorization from specified field safety officer and supervisor
- KU6.** companys reporting structure & organization culture
- KU7.** company's customer support policy
- KU8.** solar resource assessment including Direct normal irradiation, diffuse horizontal irradiation, global horizontal irradiation and albedo
- KU9.** knowledge of Excel and Solar simulation software like PV*SOL, PVsyst, etc.
- KU10.** various type of tracking system like maximum PowerPoint Tracker (MPPT) system and their usage
- KU11.** type of DC/AC pumping systems usage and their operating characteristics
- KU12.** underground water levels, availability of water and recharging frequency of water
- KU13.** efficiency, cost and typical specifications, functioning and operating principle of different types of solar photovoltaic plants, commercially available PV modules, inverters, charge controllers, battery, mounting structures, cables, junction boxes and other components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to ones role
- GS2.** read vernacular/English language

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- GS3.** read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** ability to read from different sources- books, screens in machines and signage
- GS5.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with peers, superiors and sub-ordinates
- GS10.** follow organization rule-based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize service work to meet deadlines
- GS13.** organize raw materials and packaging materials required for site survey
- GS14.** plan to utilize time and equipment's effectively
- GS15.** work constructively and collaboratively with others
- GS16.** follow code of conduct
- GS17.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS18.** recognize problems and provide solutions using a range of cognitive and practical skills
- GS19.** approach relevant authority when required
- GS20.** apply knowledge of facts, principles and processes to select the right course of action to perform tasks
- GS21.** use reasoning skills to identify and resolve basic problems
- GS22.** use intuition to detect any potential problems which could arise during operations
- GS23.** use acquired knowledge of the process for identifying and handling issues

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design, installation and commissioning of solar water pumping system</i>	26	24	-	-
PC1. discuss how to carry out Solar resource assessment	2	-	-	-
PC2. discuss how to decide on the specifications of the pumping set and motor	2	-	-	-
PC3. discuss how to decide on the capacity of PV modules	2	-	-	-
PC4. discuss how to design the plan of mounting structures and foundation	2	-	-	-
PC5. discuss how to oversee the preparation of the foundation for solar module mounting structure and motor pump set.	2	-	-	-
PC6. discuss how to ensure structure is fixed on the foundations	2	-	-	-
PC7. discuss how to oversee the mounting of solar modules; connection of solar module array to pump set in case of DC pumps and installation of inverter (in case of AC pumps).	2	-	-	-
PC8. discuss how to ensure protection system are in place	2	-	-	-
PC9. discuss how to compare the output with design output and take corrective actions, if required	2	-	-	-
PC10. discuss how to ensure connection of the solar module array to motor pump set through a Maximum Power Point Tracker (MPPT) to get maximum power from the array	2	-	-	-
PC11. discuss how to install an inverter after MPPT to convert DC power to AC power in case an AC submersible motor pump set is used	2	-	-	-
PC12. discuss how to operate solar-powered water pump systems effectively and Implement routine maintenance practices for system performance.	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss how to ensure periodical cleaning of solar module array; periodically ensure tightness of cable connections and periodic maintenance of motor pump set	2	-	-	-
PC14. show how to perform Solar resource assessment	-	2	-	-
PC15. show how to analyze the water usage and level of water table at site	-	2	-	-
PC16. show how to decide on the specifications of the pumping set and motor	-	2	-	-
PC17. show how to decide on the capacity of PV modules and design the plan of mounting structures and foundation	-	2	-	-
PC18. show how to oversee the preparation of the foundation for solar module mounting structure and motor pump set	-	2	-	-
PC19. show how to ensure that structure is fixed on the foundations	-	2	-	-
PC20. show how to perform inspection and testing of equipment	-	2	-	-
PC21. show how to perform start-up procedures and measure output	-	2	-	-
PC22. show how to compare the output with design output and take corrective actions, if required	-	2	-	-
PC23. show how to install an inverter after MPPT to convert DC power to AC power in case an AC submersible motor pump set is used	-	2	-	-
PC24. show how to assess various technical aspects of a Pump controller to perform trouble shooting of a Solar PV Pumping system	-	2	-	-
PC25. show how to operate solar-powered water pump systems effectively and implement various routine maintenance practices for improved system performance.	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	26	24	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0134
NOS Name	Design, Installation and Commissioning of Solar Water Pumping System
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	PV System Installation Engineer
NSQF Level	5
Credits	1
Version	4.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0109.Prepare a site feasibility study report	60	40	-	-	100	20
SGJ/N0146.Design of solar PV power plant	67	33	-	-	100	20
SGJ/N0132.Installation and Commissioning of solar PV power plant	67	33	-	-	100	20
SGJ/N0133.Quality Assurance of Solar PV power plant & components	25	25	-	-	50	10
SGJ/N0106.Maintain Personal Health & Safety at project site	21	29	-	-	50	10
SGJ/N0134.Design, Installation and Commissioning of Solar Water Pumping System	26	24	-	-	50	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	286	214	-	-	500	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.